LA HW

Included exercises: 7, 13, 22, 36, 39-44

Exercises 1-10

In Exercises 1–10, evaluate the determinant of each matrix using a cofactor expansion along the indicated column.

$$\begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \qquad \begin{bmatrix} 1 & -2 & 2 \end{bmatrix}$$

Original Exercise 7

$$\begin{bmatrix} 0 & 2 & 0 \\ 1 & 1 & 2 \\ 0 & -1 & 1 \end{bmatrix}$$

first column

Exercises 11-24

In Exercises 11–24, evaluate the determinant of each matrix using elementary row operations.

$$\begin{bmatrix} 0 & 0 & 5 \end{bmatrix} \qquad \begin{bmatrix} -6 & 0 & 0 \end{bmatrix}$$

Original Exercise 13

$$\begin{bmatrix} 1 & -2 & 2 \\ 0 & 5 & -1 \\ 2 & -4 & 1 \end{bmatrix}$$

Original Exercise 22

$$\begin{bmatrix} 2 & 1 & 5 & 2 \\ 2 & 1 & 8 & 1 \\ 2 & -1 & 5 & 3 \\ 4 & -2 & 10 & 3 \end{bmatrix}$$

Original Exercise 36

$$\begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & -1 & 2 & -1 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & 0 & c \end{bmatrix}$$

Exercises 39-58

In Exercises 39–58, determine whether the statements are true or false.

Original Exercise 39

ments are true or false.

The determinant of a square matrix equals the product of its diagonal entries.

Original Exercise 40

Performing a row addition operation on a square matrix

Original Exercise 41

Performing a scaling operation on a square matrix does

Original Exercise 42

Performing an interchange operation on a square matrix

Original Exercise 43

For any $n \times n$ matrices A and B, we have $\det(A + B) =$

Original Exercise 44

For any $n \times n$ matrices A and B, $\det AB = (\det A)(\det B)$.